Network Influence Maximization Project Management Report

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1 Development Methodology

This project combines both research and software development. Our primary aim was to create a set of tools to allow for research of influence maximisation across different networks, models, methods and heuristics. Using these tools we would investigate the topic in various directions attempting to create a heuristic which produces new or novel results. Following on from the Gantt chart in the progress report the development and research was split into two main stages. The project followed the phases outlined finishing the first stage on schedule.

1.1 Stage 1

The first phase of the project was focused on the software development, implementing the diffusion models, network sampling methods, known heuristics and the greedy approach alongside CELF. The problem was explored through literature review and analysis of existing research, allowing the project scope to be clearly defined, with key challenged enabled. From this I created a plan for the coding section and aimed to have it completed before the start of the second term, I accomplished this by the progress report hand in date at the start of the term.

This phase most closely resembled a traditional waterfall style methodology. Development tasks were clearly defined, expectations as to if they were met and worked as expected were easy to test. There were minimal changes required to what I was implementing, expectations and expected results were clear. The implemented code was tested by performing various trials and

experimental runs of different algorithms, testing to see if results were similar to those exhibited in the papers I had researched as well as making logical sense upon inspection.

1.2 Stage 2

The second half of the project was to utilise my code and knowledge around working with networks and the problem space to identify a new or novel solution to the influence maximization problem. This follows more closely to a typical iterative agile development methodology, implementing and quickly testing lots of different ideas to see if they are worthwhile and constantly changing the requirements based on what has worked so far and what has not. Potential ideas were implemented and explored rapidly, tested against existing methods to evaluate their effectiveness. Those methods with interesting or viable results were researched further while those which proved ineffective were discarded.

1.3 Version Control and Code Management

Version control was used throughout the project to manage the development progress and to allow me to ensure that changes to the codebase could be tracked precisely.

The project was maintained using Git, hosted on GitHub and used GitHub actions to run linting, formatting and documentation generation.

2 Time Management

At the start of the project a rough outline was made which split the project into the two described main sections. As time went on and we reached further and further into the project's timespan the goals, requirements and challenges became clearer and able to be actioned.

Weekly meetings with my supervisor were held, allowing me to talk about my progress, any issues with what I had done so far and what I might do in the next week(s) ahead. This allowed my supervisor to give valuable feedback or suggest alternative approaches that may work better than things that I have tried.

3 Stakeholder Management

A key stakeholder for this project was my supervisor. Weekly meetings were held consistently throughout the duration of the project. These meetings allowed for regular checkups of progress as well as discussing my results so far as well as approaches I have taken and would take as time progressed.

My supervisor's insight and knowledge of the topic proved particularly valuable during the second phase of the project. Having regular insight into strategies which may be a better way forward or specific insights into algorithmic approaches (particularly the community detection methods which ultimately did not pan out) helped me to avoid hyper focusing on un-viable approaches.

As the student who is being assessed on the outcome of the project I also consider myself a stakeholder. Ensuring that the project remained on track was very important to me as well as producing a high quality piece of work was dependant on my effort and taking into account any advice and feedback from my supervisor.

4 Risk Management

The primary risk associated with the project was that there would be no meaningful or significant new results found. We are studying an already dense and well researched problem, identifying a space where new contributions can be made is challenging.

4.1 Failure to find results

To mitigate this risk the research process was carefully documented throughout the project. All test scripts and notebooks were kept and version controlled. Unsuccessful ideas or thoughts were still kept and allowed me to understand the problem space better even if they don't actually produce any worthwhile results.

4.2 Loss of code or data

All base code, scripts and exploratory notebooks were stored in version control under git, backed up on github and replicated on both my personal computer and the DCS systems.

5 Issues and Challenges

There were no major unforeseen technical issues which arose during the project. However one of the main challenges that I encountered was ensuring that I remained focused, productive and on track.

5.1 Continuous Evaluation

I found it hard to explore many different ideas at the same time, becoming dis satisfied when ideas I tried did not produce viable results was a difficult challenge to deal with.

Regularly evaluating the usefulness and results produced throughout the project, including by discussing with my supervisor helped me to determine whether a path I was exploring was worth continuing or if it would be more productive to focus my attention in a another direction.

5.2 Workload Balancing

Balancing my time between a very coursework heavy second term and a part time job alongside the dissertation was a very difficult challenge. Careful scheduling of tasks and prioritization was required to ensure that I continued to make progress throughout the project.

6 Legal, Ethical, Social and Professional Issues

This is a computer science research project with no human participants involved, no collection or processing of personal data. No software is deployed or able to be used by any member of the public. The datasets are all publicly available and used extensively in academic work.

There are no professional concerns, all sources are cited appropriately, experimental results are valid and reported accurately to the best of my knowledge. No data or results have been presented to be something they are not.